

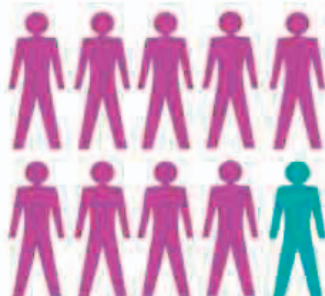


The Armagh Observatory, the oldest scientific institution in Northern Ireland, was built in 1790 and was one of the largest and most advanced observatories of its day.

IN PHILADELPHIA, AMERICAN INVENTOR OLIVER EVANS

(1755–1819) developed an idea that would make powered transport a reality. The low-pressure Watt steam engines of the day (see 1765–66) were too big and heavy for land vehicles. But Evans realized that if the steam is much hotter, the piston can be driven by steam pressure alone—so the condensing process needed to create a vacuum in low-pressure engines could be dispensed with. This revelation led to Evans' invention of **high-pressure steam engines**, which could give the same power as Watt engines that were 10 times larger.

In Warwickshire, England, inventor **John Barber** (1734–1801) was developing an even more revolutionary engine—the **gas turbine**. His idea was to compress gas made from wood or coal, then burn it explosively to turn the vanes of a paddle wheel. Barber never built a prototype, but his ideas re-emerged centuries later when the jet engine was developed. Another discovery, which is now vital for modern aircraft, was **titanium**. The metal was isolated from the mineral ilmenite in Cornwall, England, by British geologist **Reverend William Gregor** (1761–1817). It was independently discovered a few



9:1 Lightning injuries

Nine of every ten people struck by lightning survive. James Parkinson proposed that survivors suffer from a form of muscle paralysis and skin burns.

months later by German chemist **Martin Klaproth** (1743–1817), who named it titanium.

There were several developments in the field of pure science too. Swiss physicist **Pierre Prévost** (1751–1839) proved that **all bodies**—no matter how hot or cold—**radiate heat**. German physiologist Franz Joseph Gall (1758–1828) showed that the **nervous**

system is actually made from masses of nerve fibres, or ganglia. British surgeon **James Parkinson** (1755–1824), who lent his name to a devastating disease of the nervous system, gave the first medical description of injuries to people struck by lightning.

Stargazing continued to attract funding from wealthy patrons, not just because of people's fascination with it but also because of its potential value to shipping. Archbishop of Armagh, **Richard Robinson**, built an expensive **observatory in Armagh**, Northern Ireland, which continues to be one of the leading scientific research establishments today.

titanium



Titanium

The metal titanium has several ores, including ilmenite and rutile, and it also occurs in trace amounts in rocks, water bodies, soils, and all living things.



A cartoon from 1807 shows the initial mixed reaction in London to the new gas streetlighting, which would soon transform cities at night.

THIS YEAR SAW A DISCOVERY THAT WOULD ONE DAY HELP

technology transform cities across the world—**Alessandro Volta's** discovery of **chemical electricity**. Previously, static electricity had been generated through friction. Volta found that by bringing metals into contact, he could create chemical reactions that generated current electricity. He went on to produce the first battery (see 1800).

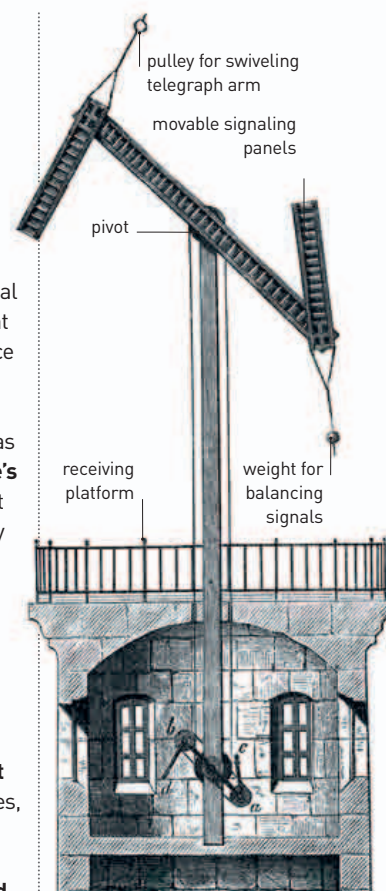
Another notable development in the field of communication was French inventor **Claude Chappe's telegraph**—a line of towers that could relay messages visually by showing panels at different angles. It could send a message across 137 miles (220 km)—from Lille to Paris—in an hour.

Scottish engineer **William Murdoch** (1754–1839) invented **gas lighting**; his home in Cornwall, England, was the **first gas-lit house**. Soon many houses, factories, and city streets would be lit by gas.

The **first iron-cased rockets**—forerunners of modern missiles—were used in India by

Tipu Sultan, ruler of Mysore, against the British.

One discovery that was little appreciated at the time was British scientist **Thomas Wedgwood's** (1771–1805) observation that all **materials**



Chappe telegraph

This telegraph tower shows Chappe's semaphore system, with its mechanism for changing the panels' angle to pass on messages.

glow the same color when heated to the same temperature. For example, we know the Sun's surface is 5,800 K (9,980°F) because of its color.

1790 American inventor **Oliver Evans** patents high-pressure steam engine

1790 British physician **James Parkinson** describes lightning injuries

1790 American inventor **Samuel Hopkins** receives the first US patent for a method of potash production

1791 British inventor **John Barber** patents the gas turbine

1791 British geologist **Reverend William Gregor** discovers titanium

1791 German physiologist **Franz Joseph Gall** describes the nervous system as masses of fibers

1791 Italian physicist **Luigi Galvani** published Commentary on the Effect of Electricity on Muscular Motion

1791 Swiss physicist **Pierre Prévost** shows that all bodies radiate heat

Italian physicist **Alessandro Volta** produces chemical electricity

British scientist **Thomas Wedgwood** notices that all materials glow the same color at the same temperature

French scientist **Claude Chappe** builds his semaphore telegraph

British engineer **William Murdoch** invents gas lighting

The first iron-cased rockets are used in Mysore, India

American engineer **James Rumford** granted English patent for water turbine